

Bale magkakaroon po kami ng dalawang system na pwedeng piliin sa software Manual mode at Ai-Assisted mode.

1. System Operation

1.1 Normal operating modes

1. Manual Mode

- The AI module sends notifications through the application for real-time monitoring alerts and detected standby loads, which the user may manually review and apply.
- Users may manually control the on or off of selected lighting and power outlets through relays, and the ACU using Infrared Radiation (IR) control via mobile application in any location.
- These actions are all logged and used by the AI module to adjust thresholds and user preferences over time for accuracy.

2. AI-Assisted Mode

- The AI module analyzes recent environmental data (occupancy, kWh, standby usage, pf) and generates suggested actions per branch, for example:
 - a. Turn OFF Outlet 2 – standby load detected during an unoccupied period of 5 minutes.
 - b. Turn ON the Living Room lighting – measured human motion from the RCWL-0516, HLK-LD2410B, and HC-SR501 PIR sensors are above the required threshold for the current occupancy condition.
 - c. Enable capacitor bank from the main panel – low power factor ($pf < 0.9$) detected on the main branch read from the PZEM-004T.
- For selected branches, AI logic can automatically execute high-confidence actions within a limited amount of time, for example:
 - a. hallway lights automatically turn OFF immediately when no motion is detected.
 - b. scheduled night-off for convenience outlets with non-critical loads, while other branches may stay in Assisted or Manual mode.
- For air-conditioning units (ACU), infrared radiation (IR) signals are used to turn ON and OFF the unit based on AI suggestions with user approval derived from the system with reference to room temperature, occupancy, and user preference.
- AI suggestions are displayed on the application through the mobile device and the mounted touchscreen monitor in the panel, with Approve and Reject buttons; actions are only executed after user approval (sound alert system that will notify users in the monitor).
- Users may override any of selected lighting and power branch, capacitor-bank state connected in the main panel, or ACU's on/off operation at any time

through the mobile application and the mounted touchscreen monitor; all overrides are automatically logged in the system.

3. Security Mode

- When the user activates Security Mode through the application dashboard, the system enhances residential safety and reduces unnecessary standby consumption while the residence is unoccupied.
- The control and automation engine automatically disconnects all non-critical convenience outlets by sending OFF commands to their corresponding relay-controlled branches, while keeping critical loads such as refrigerator, networking devices, and other protected appliances energized.
- The system also performs presence simulation by toggling selected lights during configurable evening hours based on semi-randomized schedules to create the impression that the residence is occupied.
- Security mode actions, including outlet disconnection, protected-load maintenance, and simulated lighting activity are recorded in the system logs, and may be viewed through the mobile application and mounted touchscreen monitor.
- Users may enable, disable, or override Security Mode settings at any time via application, and all overrides are automatically logged by the system.

1.2 Adaptive lighting operation (ON and OFF Operation)

- The Raspberry Pi 5 periodically acquires motion and presence data from the RCWL-0516, HLK-LD2410B, and HC-SR501 PIR sensors assigned to each area.
- The environmental data from the motion sensors are logged and sent to the main system for processing and decision-making.
- When motion is detected and the human presence is verified, the control engine:
 - Turns ON the corresponding light through a designated relay module for that specific branch.
- In Manual Mode, when no motion is detected for the configured timeout period, the system will send a notification, which the user may manually review and apply before system execution.
- In AI-Assisted Mode, when no motion is detected and the sensed data is verified for the configured timeout period, the system automatically turns it OFF immediately.
- Lighting actions, motion events, and user overrides are stored in the database for later analysis and preference learning.

1.3 Controlled convenience outlet operation

- The Raspberry Pi 5 obtains real-time electrical data from the PZEM-004T modules and occupancy information from the motion and presence sensors assigned to each area.

- For selected non-critical convenience outlet branches, the system evaluates occupancy condition, time of day, branch power level, and recent operating history before issuing any ON/OFF control action.
- In Manual Mode, the system generates notifications when no occupancy is detected for a configured period and the measured branch load indicates idle or standby behavior.
- In AI-Assisted Mode, the system may turn OFF selected non-critical outlet branches subjected to user's approval, when no motion is detected for the configured timeout and the measured branch load remains below the standby threshold for the required delay period.
- Critical loads, protected outlets, and branches assigned as always-on are exempt from automatic disconnection (e.g., outlet for refrigerator)
- All convenience outlet switching actions, user overrides, and measured standby reductions are logged for performance analysis over time.

1.4 Supplementary electrical-efficiency feature for selected inductive loads

- **Power factor monitoring and correction operation**
 - Each selected inductive branch is monitored by its assigned PZEM-004T, which measures voltage, current, active power, energy, frequency, and power factor in real time.
 - The Raspberry Pi 5 continuously reads the measured branch power factor values and stores them with timestamps in the local database for predictive learning.
 - For branches equipped with dedicated capacitor banks, the control engine compares the measured power factor against predefined thresholds such as PFmin, PFtarget, and PFmax.
 - When the branch power factor falls below the minimum threshold ($\text{pf} < 0.9$) and the branch load is sufficiently high, the system energizes the corresponding relay to connect the capacitor bank in parallel with the load.
 - When the branch power factor rises above the upper threshold or when the branch load becomes too low, the system de-energizes the relay to disconnect the capacitor bank and avoid overcompensation.
 - The system may autonomously connect or disconnect the capacitor bank for selected branches based on real-time power factor readings and confidence rules.
 - Capacitor-bank switching states, corrected power factor values, current changes, and user overrides are logged for further analyses and evaluation.

1.5 Energy monitoring and operation

- Each PZEM-004T measures voltage, current, active power, energy (kWh) and power factor for its specific branch; all 10 circuit branches have PZEM-004T, but one (1) PZEM-004T is connected to the main branch to monitor the overall consumption and power factor of all electrical loads.
- The Raspberry Pi 5 integrates power over time to monitor and record daily and monthly power (kWh) consumption.
- The energy dashboard includes and displays the following:
 - Real-time reading of Voltage (V), current (I), active power (kWh), and power factor (pf) quality per branch.
 - Overall reading of Voltage (V), current (I), active power (kWh), and power factor (pf) quality of all 10 branches.
 - Daily and monthly kWh consumption, including percentage contribution per branch.
 - Branches' ON/OFF status, capacitor-bank status, and selected control mode.
 - AI recommendations and suggestions for energy optimization based on learned user behavior and real-time monitoring data.
- The system stores all time-stamped (5-minute interval) measurements for historical analysis, anomaly detection, and AI-assisted recommendation generation.

1.6 Anomaly detection operation

- The analysis engine continuously scans logged data for:
 - Standby loads: non-zero power during periods with no motion and outside normal operating hours.
 - Overnight usage: kWh consumed above the set threshold between defined “sleep hours” (e.g., 22:00–06:00) for selected lights and convenience outlets.
 - Day-to-day anomalies: large increases in daily kWh per branch compared to recent daily averages, expected schedules, or historical usage patterns logged in the system.
 - Power factor anomalies: sustained low power factor on selected branches with inductive loads indicating the need for power factor correction.
- For each detected anomaly, the system estimates the corresponding wasted kWh consumption and generates an AI suggestion, such as “Enable auto-OFF at 23:00 for Branch 4; potential saving: 12 kWh/month.”
- In Manual Mode, anomaly-based suggestions are shown to the user for alerts, no automatic execution will occur unless the user approves the system's recommendation via application.

- In AI-assisted Mode, selected high-confidence anomaly responses may be executed automatically, subject to branch classification and safety rules. (add specific situation and appropriate mitigation)

1.7 Safety and reliability monitoring operation

The system continuously monitors branch current, branch voltage, enclosure condition, and branch response to issued control commands in order to detect abnormal or unsafe operation. For overcurrent monitoring, the Raspberry Pi 5 compares measured branch current from the PZEM-004T modules with preset allowable current limits. A warning alert is generated when the measured current reaches 80% to 99% of the branch rating for at least 10 seconds, while an overload alert is generated when the measured current reaches or exceeds 100% of the allowable current limit for at least 5 seconds.

For enclosure overtemperature detection, the system monitors the temperature inside the REMS panel or near the switching components. If the measured temperature exceeds the preset safety threshold, the system issues an overtemperature alert and records the event for maintenance review.

For actuator fault detection, the system compares issued relay commands with the resulting branch electrical behavior. If a branch is commanded OFF but significant power is still detected, or if a commanded ON action produces no expected load response, the system flags a possible relay or contactor fault.

Users may manually override any automated safety-related action through the mobile application or mounted touchscreen monitor. All safety and reliability events, user overrides, and system responses are stored in the database for evaluation and analysis.

2. Testing Procedure

2.1 Pre-commissioning checks (electrical components and wiring)

- Visual inspection
 - Verify that all MCBs, PZEM-004T modules, relay modules, contactors, capacitor banks, Raspberry Pi 5, RCWL-0516, HLK-LD2410B, HC-SR501, RTC module, and the touchscreen monitor are correctly mounted, labeled, and wired according to the single-line diagram and wiring diagrams.
 - Check proper segregation between high-voltage and low-voltage sections and ensure all covers are in place.
- Continuity and insulation
 - Confirm correct polarity (line, neutral, earth) for all circuits and connection of components.
 - Perform continuity tests on branch circuits and protective earth conductors before energizing the system.
- Power-up checks

- Energize the REMS panel with all branch MCBs initially in the OFF position.
- Verify correct DC output for the Raspberry Pi 5 power supply, sensor rails, and relay control supply.
- Boot the Raspberry Pi 5 and verify that all required services, including data acquisition, database, web server, dashboard interface, and automation services, start successfully.

2.2 PZEM-004T power measurement testing

- Reference load test
 - Connect a known reference load (e.g., 60W incandescent lamp or a small heater) to one monitored branch.
 - Compare the PZEM-reported voltage, current, and active power with a calibrated digital multimeter or plug-in energy meter.
 - Compute the percentage error and verify that it is within acceptable limits, ($\pm 2\%$ to $\pm 5\%$) to validate the acquired data.
- Multiple load levels
 - Repeat the measurement using at least one additional known load, (e.g., 100 W to 200 W), and verify consistency and stability of readings.
 - Verify that cumulative kWh increases correctly over time for a constant load.
- Per-branch verification
 - Sequentially test each PZEM-monitored branch with a similar known load to ensure correct addressing, communication, and wiring mapping (Branch ID $\leftrightarrow$ PZEM address).

2.3 Sensor and occupancy detection testing

- Motion and presence sensor testing (for lighting and power branches)
 - Simulate movement and presence in each monitored area and verify that the RCWL-0516, HLK-LD2410B, and HC-SR501 detect occupancy correctly.
 - Confirm that the occupancy or motion states are correctly transmitted to the Raspberry Pi 5 and reflected in the application.
- Sensor response verification
 - Record the response time between actual motion or presence and the detected status shown by the system.
 - Confirm that the system distinguishes occupied and unoccupied conditions according to the configured logic.

2.4 Lighting control and adaptive behavior testing

- Manual ON/OFF operation
 - Using the mobile application and mounted dashboard, manually switch each light (applied to specific lighting load only) ON and OFF and verify the corresponding lighting response.
- Sensor-triggered lighting control
 - Simulate motion in each area and verify that the correct lighting branch turns ON according to the adaptive rules involving occupancy and time of day.
- Timeout and auto-OFF
 - Leave the monitored area and ensure that the motion sensors register no motion.
 - Verify that after the configured timeout, lights then turn OFF immediately in AI-Assisted Mode.

2.5 Controlled convenience outlet testing

- Manual switching
 - Using the dashboard and mobile application, manually switch selected convenience outlet branches ON and OFF and verify correct operation.
- Occupancy and standby-based control
 - Simulate occupied and unoccupied conditions while applying a test load to the selected outlet branch.
 - Confirm that the system does not disconnect critical or protected outlets.
 - Verify that non-critical branches are suggested for disconnection in Manual Mode or automatically disconnected in AI-Assisted Mode when the occupancy timeout and standby thresholds are satisfied.
- Security Mode testing
 - Activate Security Mode and confirm that all selected non-critical convenience outlet branches are disconnected while protected loads remain energized.
 - Verify that all Security Mode actions are properly logged in the database.

2.6 Power factor monitoring and correction testing

- Baseline pf measurement
 - Connect a selected inductive load branch and record voltage, current, active power, apparent power, and power factor before capacitor-bank switching.
- Capacitor-bank switching verification
 - Use a naturally inductive load and confirm that the system detects PF below the configured threshold ($\text{pf} < 0.9$).

- Verify that the corresponding relay or contactor energizes and connects the appropriate capacitor bank for power factor correction.
- Confirm capacitor-bank connection status through auxiliary contact feedback, if implemented.
- Correction performance check
 - Record branch current and power factor before and after capacitor-bank activation.
 - Verify that power factor improves and that the capacitor bank disconnects when the load becomes too low or PF exceeds the upper threshold.
 - Confirm that overcompensation does not occur within the configured control range.

2.7 Mobile application and mounted dashboard testing

- Real-time data display
 - Open the mobile application and mounted dashboard to verify that each branch displays real-time voltage, current, power, kWh, power factor, control status, and capacitor-bank state at the expected refresh rate.
 - Confirm that daily and monthly energy values agree with manual calculations for test loads; verify also the overall energy consumption of all ten (10) branches.
- Notifications and AI suggestions
 - Create scenarios where anomaly is detected, including overnight test load operation, standby usage, or persistent low power factor.
 - Verify that the system generates clear notifications with explanations and estimated savings or corrective actions.
 - Approve or reject the generated suggestions and confirm that the corresponding actions are executed correctly and logged in the system.
- Mode switching and manual override
 - Switch between Manual, AI-assisted, and Security modes through the dashboard and verify that behavior and system operation changes accordingly.
 - Override an automatically executed action and confirm that the system respects the user command and records the override action in the database.

2.8 Safety and reliability feature testing

- Panel overtemperature testing

The temperature monitoring feature shall be verified by exposing the enclosure temperature sensor to elevated but safe temperatures and confirming that the system detects the abnormal thermal condition and generates the corresponding alert.

- Actuator fault detection testing

The actuator fault detection logic shall be tested by simulating mismatches between issued control commands and measured branch response. For example, a branch may be commanded OFF while maintaining load current, or commanded ON without the expected power draw, in order to verify that the system correctly identifies and logs the abnormal condition.

- Manual override and emergency control testing

The manual override feature shall be tested by forcing an automated or suggested action and confirming that the user can successfully cancel, replace, or override the action through the dashboard or mounted touchscreen monitor. All override actions shall be verified in the system logs.

2.9 System reliability and 24/7 operation testing

- Burn-in test
 - Operate the complete REMS system, including sensors, PZEM modules, relays, database service, and energy dashboard, continuously for at least 24 to 72 hours.
 - Monitor Raspberry Pi 5 CPU temperature, memory usage, storage status, and system logs for warnings and failures.
- Power interruption simulation
 - Safely power-cycle the system to simulate a short power outage.
 - Verify that the Raspberry Pi reboots successfully, all services restart automatically, sensors reconnect, relay states recover appropriately, and automation rules resume without manual reconfiguration.
- Data integrity check
 - Verify that energy logs before and after restart remain consistent and that no significant database corruption or data loss has occurred.

SUMMARY:

Basically, ang utak ng lahat dito ay yung Raspberry Pi 5. Siya yung magsisilbing local server, magpapakabo ng database, at hahawak ng automation engine natin. Ang trabaho ng hardware layer ay magbasa ng data at mag-execute ng commands. May labing-isang PZEM-004T modules tayo—sampung piraso para sa bawat branch circuit, tapos isa para sa main line para sa kabuuang V, I, W, kWh, at PF readings. Tapos, kada limang minuto, ire-record ng Pi 5 lahat ng data na 'to sa DB. Para naman sa occupancy, pinagsama-sama natin yung RCWL-0516 radar, HLK-LD2410B mmWave, at HC-SR501 PIR sensors para siguradong accurate kung may tao ba talaga sa kwarto o wala. Direct control naman tayo sa relays, contactors, at may IR blaster tayo na nakatutok para sa ACU. Lahat ng estado nito, dapat nagsync real-time sa mobile app at sa mounted touchscreen natin.

Pagdating sa software logic, hahatiin natin 'to sa tatlong modes na pwedeng lipat-lipatan ng user. Sa **Manual Mode**, parang dashboard lang ang app—nagmomonitor lang at nagsend ng notifications kung may napansing aksaya, pero strictly walang galawan ng

relays o IR commands hangga't walang pinipindot yung user. Sa **AI-Assisted Mode**, dito na papasok yung automation: mag-a-analyze ang engine ng sensor data at maglalabas ng suggestions na may "Approve" at "Reject" buttons sa UI. Pag high-confidence action halimbawa, hallway lights na naconfirm talagang walang tao pagkatapos ng timeout pwede nating i-auto execute agad. Pero take note, kapag ACU control, kinakailangan pa rin ng explicit click sa Approve button bago mag-fire yung IR signal. Huli sa modes ay ang **Security Mode** para kapag walang tao sa bahay; matic nitong papatayin lahat ng non-critical outlets habang sinisiguradong nananatiling nakasaksak at buhay ang mga critical loads tulad ng refrigerator at router. Magpapatakbo rin 'to ng semi-random lighting patterns sa gabi para mag-simulate na may tao sa loob.

May dalawa tayong special sub-routines na kailangan mong i-code nang maayos. Una, yung **Adaptive Lighting at Outlets**: kapag may na-detect na tao, magoON ang ilaw. Kapag nag-timeout na walang motion, mag-e-execute agad ng OFF sa AI Mode, o magpapadala lang ng prompt sa Manual Mode. Sa outlets naman, ie-evaluate ng system kung standby load lang ba yung nakasaksak at walang tao—kapag oo, mag-a-suggest din 'to na i-cutoff yung branch, basta siguraduhin mong nakabukod at "exempted" sa logic yung mga critical outlets. Pangalawa, yung **Power Factor Correction**: continuous mong babasahin yung PF sa PZEM ng inductive branches. Kapag bumaba sa $PF < 0.9$ habang may load, ima-manually trigger ng Pi yung relay para ikabit yung capacitor bank. Kapag maayos na uli ang PF o kapag sobrang baba na ng load, ide-deenergize mo rin agad para maiwasan natin yung overcompensation (pagnakaopen yung aircon automatic mamatay ang capacitor bank).

Para naman sa analytics at safety, gagawan mo 'to ng background thread para sa anomaly engine. Mag-i-scan 'to ng DB logs para mag-flag ng standby leaks, overnpute mo lang yung estimated monthly waste na ipapakita sa user. Sa safety monitoring, magse-set ka ng threshold checks: kapag umabot sa 80%–99% ng rating yung branch current sa loob ng 10 seconds, mag-trigger ka ng warning alert; kapag umabot o lumagpas sa 100% sa loob ng 5 seconds, overload alert na agad. Maglagay ka rin ng actuator fault logic kung saan ikokompara mo yung relay command sa totoong power draw—halimbawa, nag-command ka ng OFF pero may dumadaloy pa ring kuryente, ibig sabihin may sira yung relay.

Sa testing at verification natin, ang kailangan mo lang i-check: siguraduhing accurate yung PZEM readings kumpara sa physical multimeter at siguraduhing auto-restart at nagse-save ng DB state ang Pi kapag biglang na-unplug o namatayan ng kuryente, at patakbuhan mo lang ng 24 to 72 hours na burn-in test para masigurong walang memory leak o system crash.

Materials
Microcontrollers
Raspberry Pi 5
Monitor
7in Monitor Touchscreen
Sensors
PZEM-004T AC Power Monitor (11 pcs)
RTC Sensor
HLK-LD2410B
IR Sensor
Actuators & Load Control
Contactors
Relay Module 2-ch
Relay Module 1-ch (20 pcs)
Capacitor bank



REMS
Residential Energy Management System
with Power Factor Correction

 Dashboard

 Energy Monitoring

 Power Factor Analysis

 Load Control

 AI Recommendations

 Historical Reports

 Alerts & Notifications

 Settings

 About System

System Status

Online

August 06, 2026

5:25 pm EST

 DASHBOARD

  Admin 

Total Power

Energy Today

Estimated Bill

Power Factor (pf)

POWER FACTOR (PF)

POWER FACTOR TREND (TODAY)

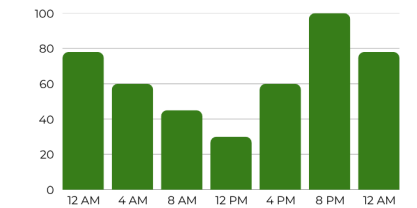
QUICK STATUS

ENERGY CONSUMPTION (kWh)

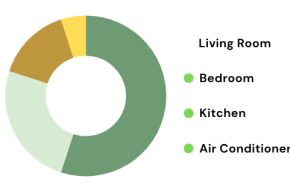
Daily

Weekly

Monthly



LOAD DISTRIBUTION





REMS
Residential Energy Management System
with Power Factor Correction

Dashboard

Energy Monitoring

Power Factor Analysis

Load Control

AI Recommendations

Historical Reports

Alerts & Notifications

Settings

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LOAD CONTROL



REMS
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Load Control

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August 06, 2026

5:25 pm EST

AI RECOMMENDATIONS

Recommendations

Insights

Alerts

Recommendation	Priority	Impact	Action
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REMS

Residential Energy Management System

with Power Factor Correction

Dashboard

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HISTORICAL REPORTS

Admin

Energy

Power Quality

Power Factor

Cost

PFC Operations

Select Range

This Month

Generate Report



REMS

Residential Energy Management System

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AI Recommendations

Historical Reports

Alerts & Notifications

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5:25 pm EST

ALERTS & NOTIFICATIONS

Admin

Recommendations

Insights

Alerts

Alerts / Notifications

Severity

Time

Status

Action

**REMS**
Residential Energy Management System
with Power Factor Correction

Dashboard

Energy Monitoring

Power Factor Analysis

Load Control

AI Recommendations

Historical Reports

Alerts & Notifications

Settings

About System

System Status

Online

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SETTINGS

General

PFC Settings

Billing

Users

Systems

SYSTEM SETTINGS

System Name

REMS

Location

Rojas Residence

Time Zone

(GMT+08:00) Asia, Philippines

Date & Time

August 7, 2026 4:04 PM

Sync

SAVE CHANGES

PFC SETTINGS (AUTO-CORRECTION)

Target Power Factor

0.95

Minimum PF Limit

0.90

Capacitor Switching Delay

30 sec

Reconnection Delay

60 sec

Capacitor Bank Sequence

C1 → C2 → C3

SAVE CHANGES

Note: dapat may tab din na nakikita kung ano mga nakabukas